

ARUN ARUNACHALAM

Jersey City, NJ | 908-330-1050 | asnarun75@gmail.com | linkedin.com/in/arunarunachalam75

PROFESSIONAL SUMMARY

Generative AI and platform engineering leader with 17+ years of end-to-end solution delivery across capital markets and enterprise data systems, including hands-on experience building production-ready GenAI platforms using RAG pipelines and agentic AI frameworks. Proven track record translating AI strategy into scalable, deployment-ready capabilities that drive measurable operational efficiency and business value across mission-critical financial environments. Skilled at managing the full GenAI project lifecycle — from ideation and scoping through deployment and post-launch support — while aligning cross-functional stakeholders across Data, Cloud Architecture, MLOps, and business units. Combines deep technical expertise in LLMs, cloud-native architecture (AWS, Kubernetes), and MLOps best practices with executive-level program management and governance leadership.

SIGNATURE IMPACT

- **\$8M** enterprise transformation → **40% performance improvement, 30% cost reduction**
- **AI-assisted engineering** → **35% faster delivery, 45% productivity gain**
- Scaled platforms from 10 to **50+ systems** with **99.97% uptime**
- Enabled **~\$2B capital efficiency** via risk platform transformation

CORE CAPABILITIES

GenAI Platforms (RAG Pipelines, Agentic AI, LLMs) | MLOps & AI-Assisted SDLC | Cloud-Native Architecture (AWS, Kubernetes) | Event-Driven Systems (Kafka) | LangChain / LlamaIndex / Autogen Frameworks | Ethical AI Governance & Compliance | Data Engineering (Spark, Databricks, ETL/ELT) | CI/CD & Agile Delivery | Executive Stakeholder & Program Management | Global Team Leadership | Trading & Risk Platforms | Regulatory Compliance (FRB, FINRA, SEC)

PROFESSIONAL EXPERIENCE

Bank of America — Director, Global Risk & Capital Markets Technology (2015 - Present)

- Architected and delivered production-ready GenAI platforms leveraging RAG pipelines and agentic AI systems, driving **45% productivity gain** and **35% faster delivery** across engineering workflows
- Implemented AI-assisted SDLC workflows and ML-based anomaly detection, embedding MLOps best practices and continuous improvement into the delivery lifecycle
- Led end-to-end platform transformation across trading and data systems supporting **\$1T+ daily processing**, managing full project lifecycle from ideation and scoping through deployment and post-launch support
- Modernized legacy systems into cloud-native, event-driven architecture (AWS, Kafka, Kubernetes), enabling scalable, high-availability infrastructure with **99.97% uptime**
- Delivered real-time trading, pricing, and risk platforms improving throughput by **40%**

- Served as primary stakeholder point of contact for enterprise AI and platform programs; managed expectations, communicated progress, and identified and mitigated delivery risks across global teams
- Collaborated cross-functionally with cloud architecture, data engineering, and business unit teams to align GenAI delivery roadmaps with strategic business objectives
- Built and led global engineering teams (**30+ across US/UK/India**), fostering a culture of execution, collaboration, and continuous improvement
- Enforced regulatory compliance and governance frameworks (FRB, FINRA, SEC) across mission-critical AI and data systems, ensuring ethical and responsible deployment standards
- Drove adoption of CI/CD practices, Agile/Scrum methodologies, and software engineering best practices across distributed teams

Bank of America — VP, Structured Products & Mortgage Technology (2011 - 2015)

- Led consolidation of risk systems contributing to **~\$2B capital efficiency**
- Built real-time pricing and analytics platforms for structured products and mortgage trading
- Managed cross-functional program delivery across data, risk, and front-office technology teams

Bank of America — AVP, Whole Loan Trading Desk Support (prior to 2011)

- Architected and delivered end-to-end Java EE trading support applications
- Introduced CI/CD practices reducing release cycle from monthly to weekly

Senior Programmer Analyst — Knight Trading (2004)

DB2 / WebSphere Administrator & Sr. Java Developer — Kyocera Mita (2003)

Sr. Java Web Developer — Mellon Investor Services (2001)

Sr. Java GUI Developer — Thomson Financials (2001)

Sr. Java GUI Developer — Icor Brokerage (2000)

AI & INNOVATION

- Building production-ready GenAI platforms using RAG pipelines and agentic AI systems (LangChain, LlamaIndex, Autogen) for real-world enterprise applications
- Applied ML for anomaly detection and operational intelligence across large-scale financial data systems
- Driving AI-assisted SDLC workflows with embedded MLOps practices, improving delivery speed, quality, and governance
- Implementing ethical AI and governance frameworks to ensure responsible, compliant deployment of generative AI capabilities

EDUCATION

Bachelor of Technology — Anna University

SKILLS

Leadership: Executive stakeholder management, global team leadership, P&L ownership, strategic planning, program management, Agile/Scrum, ITIL, SOX, FRB/FINRA/SEC compliance, ethical AI governance **AI & GenAI:** RAG pipelines, agentic AI, LLMs, LangChain, LlamaIndex, Autogen, ML anomaly detection, AI-assisted SDLC, MLOps, NLP, Hugging Face **Architecture:** Cloud-native architecture (AWS, Kubernetes), enterprise architecture, event-

driven systems (Kafka), distributed systems, high-availability platforms (99.97% uptime), Docker, containerization **Data Engineering:** Spark, Databricks, ETL/ELT, Snowflake, Hadoop, Cassandra, MongoDB **Engineering:** Java, Python, Node.js, SQL, Oracle, Sybase, REST/SOA, WebLogic, Kafka, Kubernetes, AWS **Tools:** ServiceNow, JIRA, Git, Visio, Microsoft Project